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Lecture #158

Acute Kidney Injury: Clinical Findings, Diagnosis, and Treatment





Megha Schmalzle, Academic Medicine Scholar
& Paula Ryo, D.O.

NYIT College of Osteopathic Medicine
Email: mschma01@nyit.edu, pryo@nyit.edu

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Learning Objectives

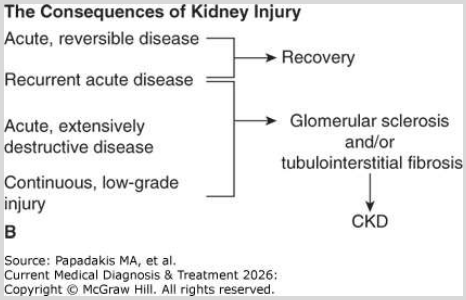
1. Define acute kidney injury (AKI) and describe its general clinical presentation
2. Differentiate the prerenal, intrinsic, and postrenal etiologies of AKI
3. Determine the appropriate workup (physical findings, lab tests, imaging) for accurate diagnosis of prerenal, intrarenal, and postrenal AKI
4. Identify the potential complications associated with AKI
5. Develop a clinical management/treatment plan for a patient with AKI

What is Acute Kidney Injury (AKI)?

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Acute kidney injury = an abrupt decline in glomerular filtration rate (GFR)

- How is this measured?
  - Hallmark = azotemia
  - Oliguria
- AKI refers to a *group* of conditions that share common outcomes:
  - Sudden loss of renal function
  - Azotemia – retention of nitrogenous wastes
  - Oftentimes the rapid dysregulation of electrolytes, acid-base, and fluid imbalances
- Depending on the cause, injury may reverse completely, partially, or worsen
- AKI is best distinguished from chronic kidney disease with medical record review



Azotemia = retention of nitrogenous wastes that causes increased BUN and Cr conc
Oliguria = decreased urine output
Mention: AKI used to be called ARF, it's the same thing

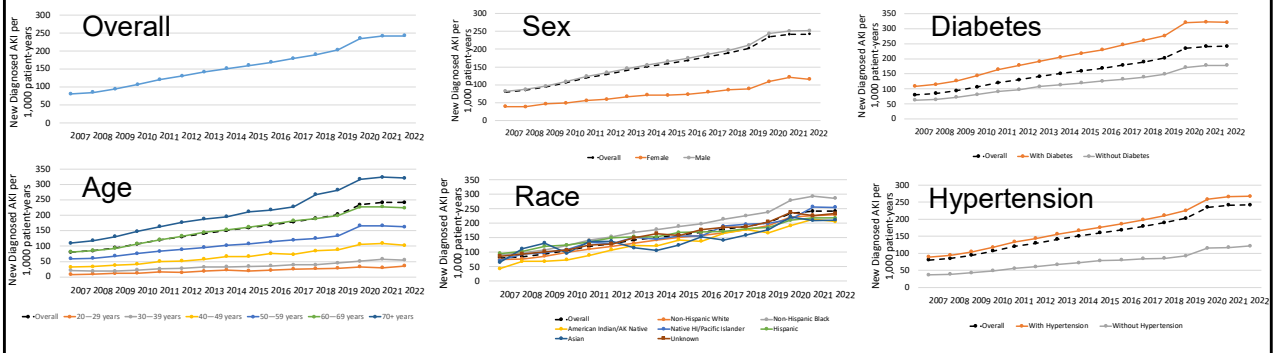
Figure B: Narrowing the differential diagnosis of kidney disease to a structural compartment can be helpful

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Epidemiology

In the United States, it's estimated that ~1% of all patients admitted to a hospital present with AKI, while 2-5% will develop it during hospitalization. This percentage rises up to 50% in ICU patients, given the severity of their disease.

Trends in Incidence Rate of Acute Kidney Injury by Diagnosis Code



The incidence rate of newly diagnosed AKI has increased from 80 per 1,000 patient-years in 2007 to 242 per 1,000 patient-years in 2022. The rise in AKI incidence after 2019 may be related to COVID-19 infections during the pandemic and then levels off. This analysis included Veterans with at least one AKI hospitalization.

Data Source: NATIONAL VA (obtained from CDC website)

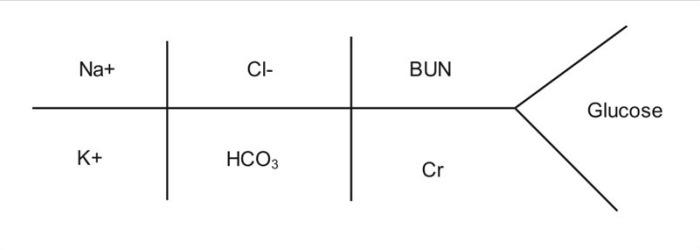
How to Interpret This Graph

If the rate in 2021 is around 250 per 1,000 patient-years, it means:
For every 1,000 patients followed for 1 year, 250 new cases of AKI were diagnosed.
This is equivalent to saying: A 25% annual incidence of AKI in that population or 1 in 4 patients developed AKI per year in that group

Diagnostic Criteria of AKI:

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1. Increase in serum creatinine of ≥0.3 mg/dL within 48 hours
2. Increase in serum creatinine to ≥1.5 times baseline within 7 days
3. Urine volume <0.5mL/kg/h for 6 hours



Note: These criteria are based on The Kidney Disease: Improving Global Outcomes (KDIGO) definition, which is the most recent and preferred

Criteria for AKI is 1 OR 2 OR 3

Normal Creatinine: 0.7 to 1.3 in males and 0.6 to 1.1 in females

Verbally mention: a 60 kg woman (~132 lbs) should make at least 30mL of urine per hour (which is equivalent to 2 tablespoons or 1 fluid oz)

Urine Output in AKI

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- Not all patients with AKI have decreased urine output
- Many patients are non-oliguric; they maintain normal urine output (>400 mL urine output per day) even if their GFR is decreasing
- Oliguria (<400 mL urine output per day) is an indicator of a worse prognosis. The lower the urine output, the greater risk for volume overload and hyperkalemia

As define by KDIGO guidelines, AKI can be diagnosed by either an increase in serum Cr or a oliguria, and these criteria are considered additive and independent. Many patients with AKI are identified solely by changes in serum creatinine

without meeting oliguria criteria, and conversely, some patients meet AKI criteria based only on urine output. Studies in critically ill populations have shown poor agreement between serum creatinine and urine output criteria, with a substantial proportion of AKI cases identified exclusively by one or the other. Nonoliguric AKI is well recognized, particularly in cases where tubular concentrating ability is impaired, such as in certain intrinsic renal diseases or sepsis, and urine output may

remain normal despite significant renal dysfunction.

Verbally mention: anuric AKI is <100 mL urine output per day

Common Clinical Findings

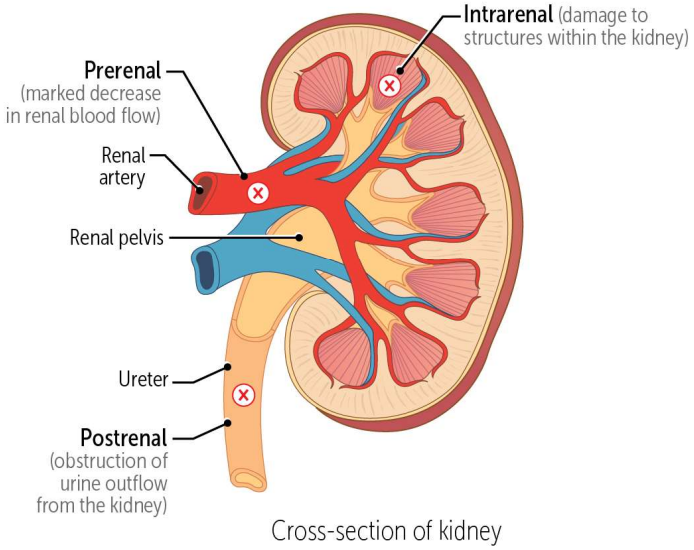
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- Increasing creatinine (Cr) in itself does not directly cause symptoms. It may be an incidental finding while ordering blood work for other conditions
- Oliguria or anuria → peripheral and/or pulmonary edema (signs of volume overload)
- Symptoms relating to underlying cause of AKI
  - Volume-depleted patients may present with vomiting and diarrhea
  - Patients with urinary retention may have suprapubic pain and fullness
- Symptoms of severe AKI with marked reduction of GFR: confusion, dyspnea, chest pain, nausea (uremic symptoms specific to renal failure)

Normal GFR 90-120. <15 mL/min is renal failure or end-stage renal disease

Why does GFR decline suddenly?

There are 3 basic mechanisms of kidney injury.

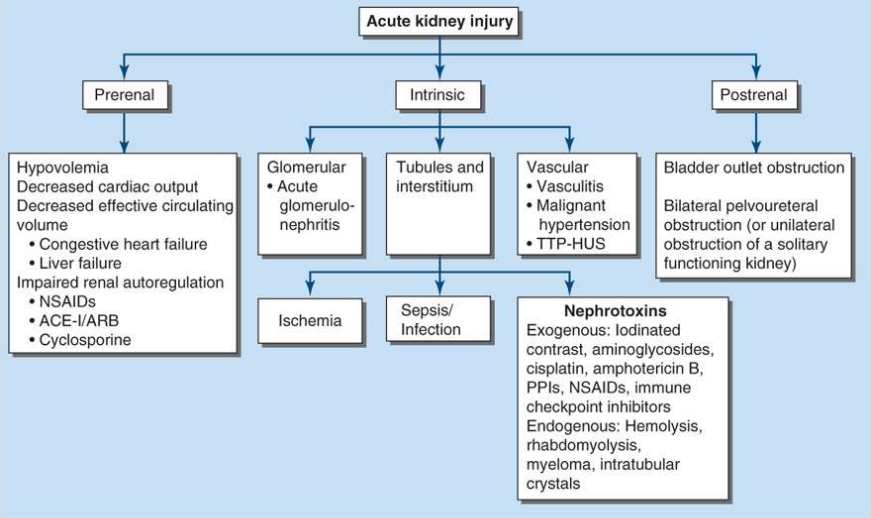


Cross-section of kidney

Reiterate that acute kidney injury is defined by a decline in GFR, but there are many different diagnoses that decrease GFR. AKI refers to a collection of diagnoses based on the mechanism of injury. There are 3 main mechanisms of injury in AKI.

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Classification of the major causes of acute kidney injury.



Source: Longo D, Fauci A, Kasper D, Hauser S, Jameson JL, Loscalzo J, Holland S, Langford C: Harrison's Principles of Internal Medicine, 22nd Edition
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ACE-I, angiotensin-converting enzyme inhibitor-I; ARB, angiotensin receptor blocker; NSAIDs, nonsteroidal anti-inflammatory drugs; PPI, proton pump inhibitors; TTP-HUS, thrombotic thrombocytopenic purpura–hemolytic-uremic syndrome.

Prerenal acute kidney injury

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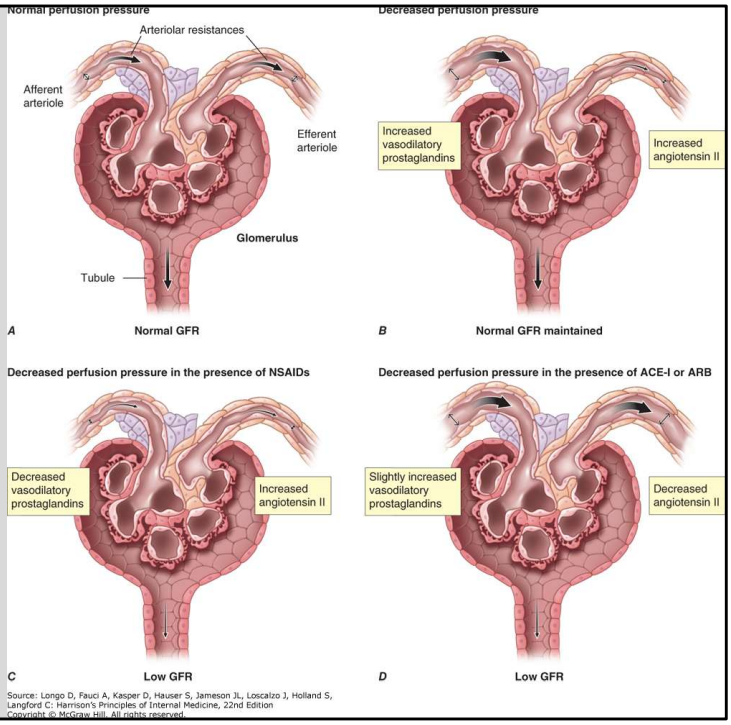
Mechanism of prerenal AKI = *decreased blood flow perfusing the kidney*

May be due to…

1. **Hypovolemia** – most common cause of prerenal AKI
   - volume depletion (hemorrhage, diuretics, diarrhea, dehydration) which leads to low renal blood flow
   - correct with volume repletion (oral fluids, IV normal saline)
2. Hypotension
   - When BP drops, autoregulation fails and GFR falls
   - Ex: shock, vasodilating drugs, hepatorenal syndrome
3. Decreased cardiac output (e.g., CHF, valvular disease, pulmonary HTN)
4. Acute blockage/narrowing of renal arteries (e.g., blood clots)
5. Afferent arteriolar constriction or efferent arteriolar dilation – both reduce glomerular capillary hydrostatic pressure and GF

Prerenal acute kidney injury

Intrarenal mechanisms for autoregulation of the glomerular filtration rate (GFR) under decreased perfusion pressure and reduction of the GFR by drugs. *A.* Normal conditions and a normal GFR. ***B.*** Reduced perfusion pressure within the autoregulatory range. Normal glomerular capillary pressure is maintained by afferent vasodilatation and efferent vasoconstriction. Angiotensin constricts afferent (preglomerular) and efferent (postglomerular) arterioles but preferentially increases efferent arteriolar resistance. ***C.*** Reduced perfusion pressure with a nonsteroidal anti-inflammatory drug (NSAID). Loss of vasodilatory prostaglandins increases afferent resistance; this causes the glomerular capillary pressure to drop below normal values and the GFR to decrease. ***D.*** Reduced perfusion pressure with an angiotensin-converting enzyme inhibitor (ACE-I) or an angiotensin receptor blocker (ARB). Loss of angiotensin II action reduces efferent resistance; this causes the glomerular capillary pressure to drop below normal values and the GFR to decrease.



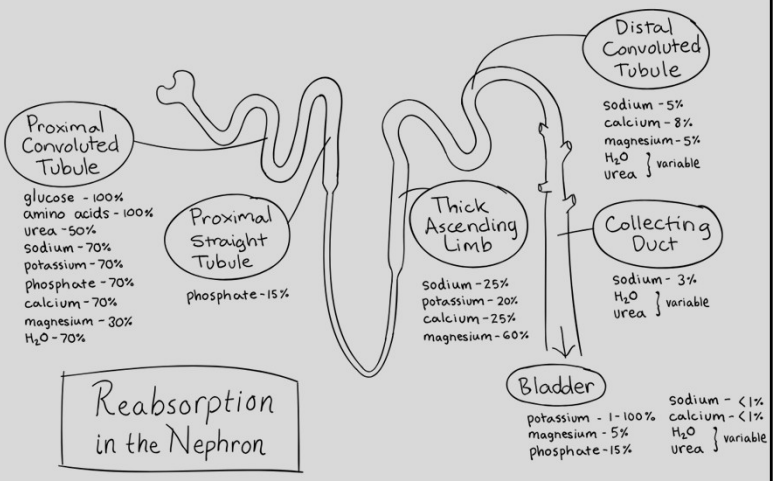
(From JG Abuelo: Normotensive ischemic acute renal failure. N Engl J Med 357:797, 2007. Copyright © 2007, Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.)

Prerenal acute kidney injury

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Serum BUN:Cr ratio >20

Decreased renal perfusion (e.g., from hypovolemia, heart failure) leads to increased reabsorption of sodium and water in the proximal tubule. Urea, which is passively reabsorbed along with water, is reabsorbed to a greater extent while creatinine reabsorption remains minimal.



Note: the diagnostic utility of the BUN:Cr ratio is limited. This finding is neither sensitive nor specific for prerenal causes and must be interpreted in clinical context.

In contrast, in intrinsic AKI (such as acute tubular necrosis), tubular injury impairs the reabsorption of urea, so both BUN and creatinine rise in parallel → the ratio remains closer to normal (10–15:1)

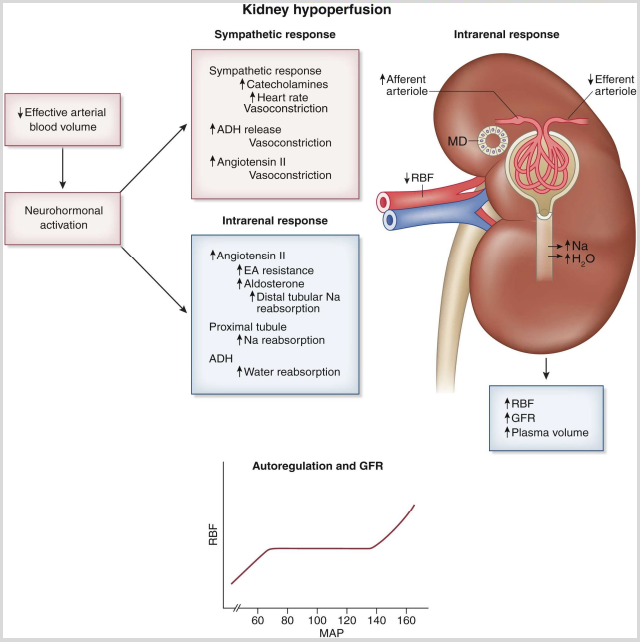
Photo source: Khan Academy

Prerenal acute kidney injury

Prerenal AKI is rapidly reversible if treated early!

The body initiates compensatory mechanisms in response to kidney hypoperfusion in an attempt to normalize RBF and GFR:

If prolonged, inadequate perfusion can lead to ischemia and potentially acute tubular necrosis.



Catecholamines increase heart rate and vasoconstriction to increase BP, antidiuretic hormone (ADH) release increases to induce vasoconstriction, and angiotensin II production also causes vasoconstriction. This results in increased CO and BP. Within the kidney, macula densa (MD)–mediated angiotensin II release increases efferent arteriole resistance, leading to stabilization of filtration pressure and GFR, and production of aldosterone, resulting in higher Na reabsorption to increase plasma volume. ADH also increases water reabsorption to increase plasma

volume in conjunction with the increased Na reabsorption. This results in an increase in plasma volume and maintenance of GFR. The lower graph shows autoregulation of RBF and GFR mediated by the intrarenal measures in response to reduced RBF. Autoregulation of RBF and GFR occurs between a mean arterial pressure (MAP) of 70 and 140 mm Hg. EA, efferent arteriole.

Source: Molitoris, B. Low-Flow Acute Kidney Injury. CJASN, 17: 1039, 2022. *DOI:* 10.2215/CJN.15341121

Intrarenal / intrinsic acute kidney injury

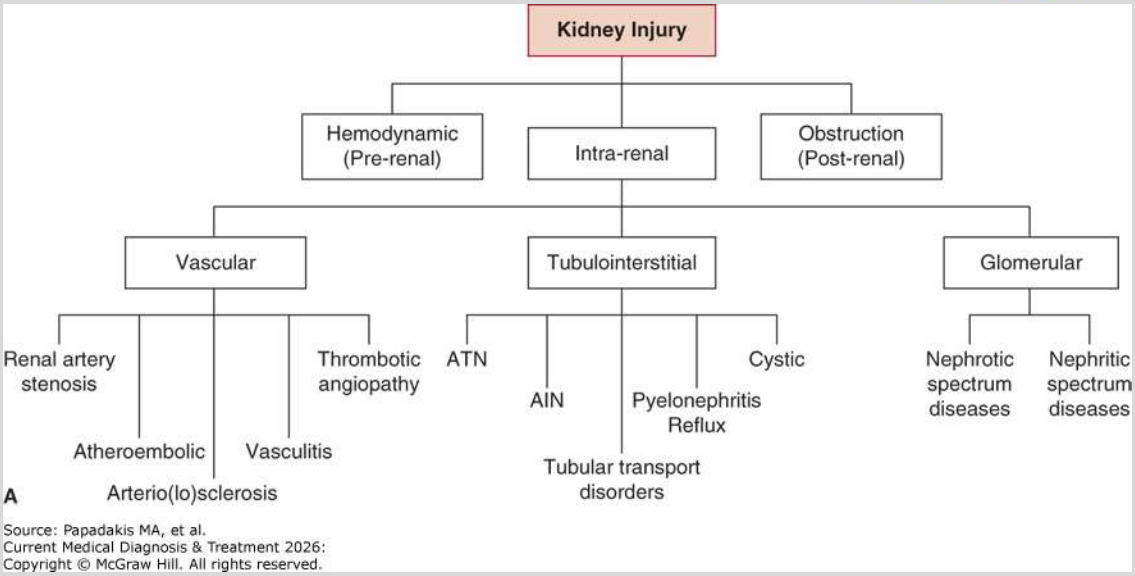
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Mechanism of intrarenal AKI = *disease within the kidneys*

- Classified by site of injury:
  - Vascular
  - Glomerular
  - Tubules & interstitium
- BUN/Cr ratio is preserved (normal 10:1 to 20:1)
- The most common causes of intrinsic kidney injury in hospitalized patients include acute tubular necrosis (ATN) and acute interstitial nephritis (AIN)

Intrarenal / intrinsic acute kidney injury

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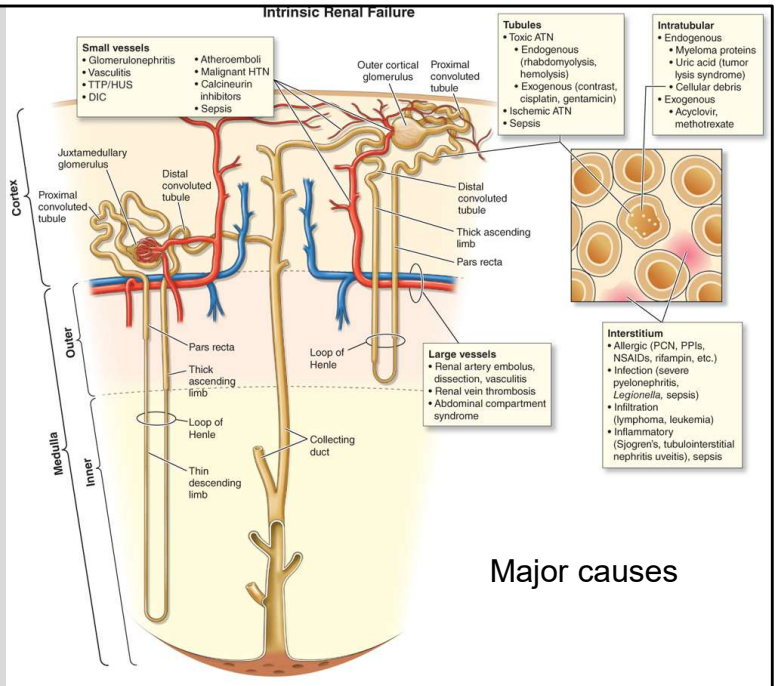


Source: Papadakis MA, et al.
Current Medical Diagnosis & Treatment 2026:
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Kidneys may be damaged by a variety of insults/disease states.

Intrarenal / intrinsic acute kidney injury

ATN, acute tubular necrosis; DIC, disseminated intravascular coagulation; HTN, hypertension; PCN, penicillin; PPI, proton pump inhibitors; TINU, tubulointerstitial nephritis-uveitis; TTP/HUS, thrombotic thrombocytopenic purpura/hemolytic-uremic syndrome.



Intrarenal / intrinsic AKI: **Acute Tubular Necrosis**

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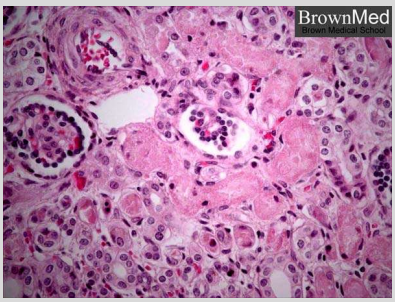
- Necrosis of renal tubular cells
- Causes:
  - Ischemic → 2° to decreased renal blood flow (e.g., prerenal azotemia)
  - Nephrotoxic → e.g., radiocontrast, vancomycin, aminoglycosides, cisplatin, myoglobinuria, hemoglobinuria
- Proximal tubules and thick ascending limb are particularly susceptible to injury
- Clinical features:
  - Elevated BUN and creatinine
  - Oliguria
  - Hyperkalemia
  - Urinalysis: + granular casts, - protein and blood
  - Classic urine microscopy finding = muddy brown casts

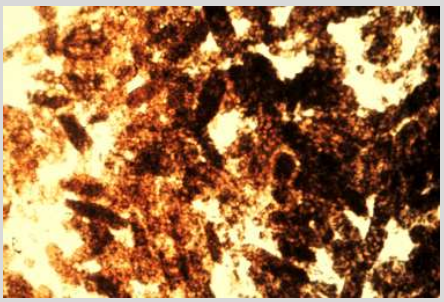
Oliguria due to the back-leak of tubular fluid around damaged tubular cells, rather than forming urine
Hyperkalemia (due to decreased renal excretion) with metabolic acidosis

Intrarenal / intrinsic AKI: **Acute Tubular Necrosis**

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- Onset 24-48 hours with 5 to 7 day duration (spontaneously resolves in many cases). Supportive dialysis is necessary if electrolyte imbalances are severe.
- Oliguria can persist for 2-3 weeks before recovery because the tubular cells take time to reenter the cell cycle and regenerate.





"muddy brown" casts

Pathology:

Photomicrograph of urine sediment showing multiple muddy brown granular casts. These findings are highly suggestive of ATN in a patient with AKI.

Intrarenal / intrinsic AKI: **Acute Interstitial Nephritis**

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- Acute inflammation of the renal interstitium
- Causes:
  - Drugs (~70-75% of cases) – NSAIDs, penicillins, cephalosporins, rifampin, sulfonamides, ciprofloxacin, loop diuretics, PPIs, allopurinol, 5-ASA, immune checkpoint inhibitors
  - Autoimmune disease (e.g., Sjogren syndrome, SLE, sarcoidosis)
  - Infection (e.g., mycoplasma)
- Generally takes 5-7 days of drug exposure to occur

Mneumonic → **5 P'S**

Painkillers (NAIDs)
Pee pills (diuretics)
Penicillins, cephalosporins
Proton pump inhibitors
RifamPin
Sulfa drugs

Intrarenal / intrinsic AKI: **Acute Interstitial Nephritis**

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- Diagnostic evaluation:
  - May be asymptomatic!
  - Fever, rash, pyuria, hematuria, CVA tenderness
  - UA → + microscopic hematuria, + WBC w/o bacteria, + eosinophils
    Note: The presence of WBC casts on urine microscopy is highly suggestive of AIN, but these are absent in the majority of AIN cases. Evaluation for urine eosinophils is no longer recommended due to lack of diagnostic utility.
  - Kidney biopsy is confirmatory
- Primary treatment = discontinue offending agent(s)

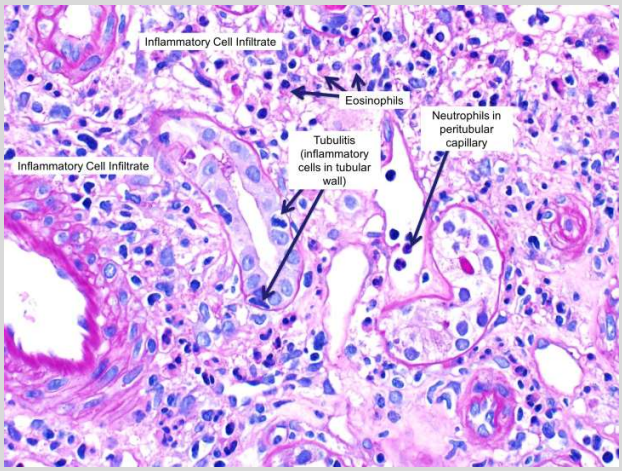


Photo source: https://nephsim.com/case-2-diagnosis-conclusions/

Postrenal acute kidney injury

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Mechanism of postrenal AKI = *obstruction to renal outflow into the collecting system*

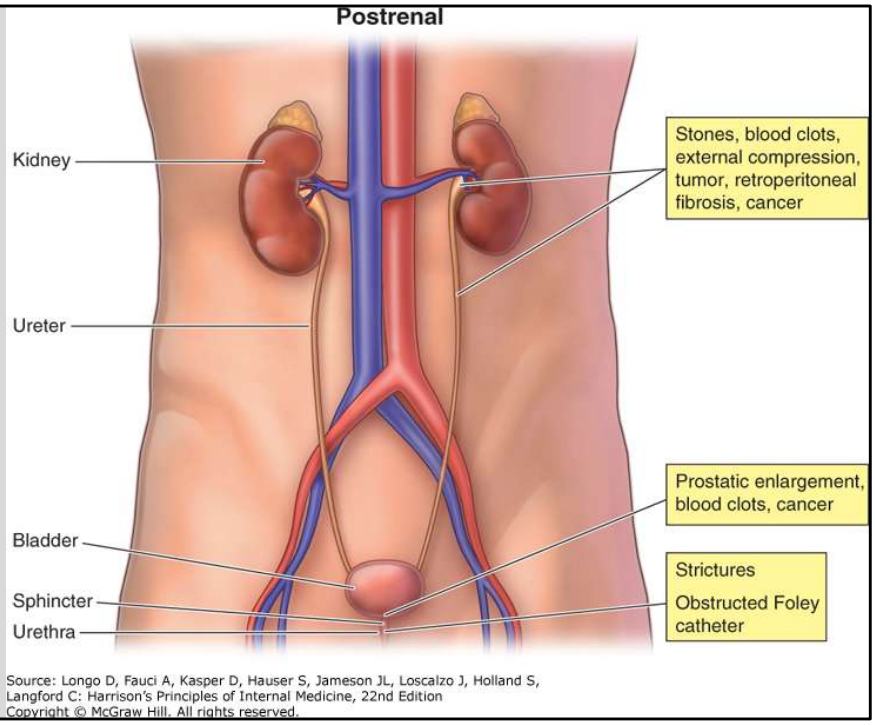
Obstruction can occur from pelvis of kidney to urethra → decreased GFR

Examples:

- Benign prostate hyperplasia (BPH) – an enlarged prostate gland can decrease bladder drainage and increase pressure in the ureters and renal tubules, reducing GFR. Most common cause of postrenal AKI in males.
- Renal stones
- Blood clots in ureters
- Cancer – bladder carcinoma, pelvic tumors (including uterine and ovarian tumors in females)
- Bladder outlet obstruction (spinal cord lesions)
- Vesicoureteral reflex in children

Postrenal acute kidney injury

Anatomic sites and causes of obstruction leading to postrenal AKI



Source: Longo D, Fauci A, Kasper D, Hauser S, Jameson JL, Loscalzo J, Holland S, Langford C: Harrison's Principles of Internal Medicine, 22nd Edition
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Postrenal acute kidney injury

- Testing considerations → ultrasound for hydronephrosis, urinary catheterization can be diagnostic and therapeutic
- If obstruction is suspected, evaluate and treat quickly. Duration of obstruction is inversely correlated with recovery of renal function.
- If obstruction is superior to the level of the bladder, it will only cause obstruction to one kidney and the serum creatinine may not change (due to hyperfiltration of the other normal kidney)



Ultrasound showing normal kidney

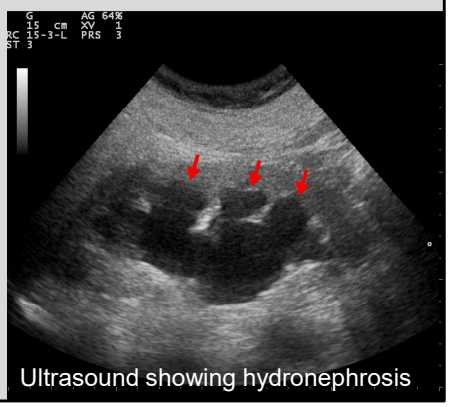

Ultrasound showing hydronephrosis

Photo source: Scholar Rx: Acute Kidney Injury

Diagnostic Evaluation of AKI

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- AKI is a *clinical* diagnosis. H&P is important!
  - Does the patient have a baseline serum Cr and GFR?
  - Medication reconciliation
- Workup includes:
  - Blood tests → CBC, CMP, BUN/Cr, uric acid, PT/PTT, CK, myoglobin
  - Urinalysis
  - Sodium load that is not reabsorbed = fractional excretion of sodium (FENa).
    - FENa = $(Na_{urine} \times Cr_{serum} / Na_{serum} \times Cr_{urine}) \times 100$
    - Note: not always a reliable test, influenced by salt-heavy diet and ongoing diuretic use

How can AKI can be differentiated from CKD? Serial blood tests showing a continued substantial rise of SCr (≥0.3 in 48 hours or ≥1.5 times baseline in 1 week) represent clear evidence of AKI. CKD is characterized by an estimated GFR <60 mL/min for at least 3 months.

The distinction between AKI and CKD is straightforward when a recent baseline SCr concentration is available, but more difficult in the many instances in which the baseline is unknown. In such cases, clues suggestive of CKD can come from radiologic studies (e.g., small, shrunken kidneys with cortical thinning on renal ultrasound) or lab tests (e.g., secondary hyperparathyroidism with hyperphosphatemia and hypocalcemia).

No set of tests, however, can rule out AKI superimposed on CKD because AKI is a frequent complication in patients with CKD, further complicating the distinction.

Once the diagnosis of AKI is established, its cause needs to be determined because the elevation of SCr or reduction in urine output can be due to a large number of physiologic and pathophysiologic processes, as described previously.

Staging of Acute Kidney Injury Severity

Stage	Serum Creatinine	Urine Output
1	1.5–1.9 times baseline OR ≥0.3 mg/dL (≥26.5 µmol/L) increase	<0.5 mL/kg per h for 6–12 h
2	2.0–2.9 times baseline	<0.5 mL/kg per h for ≥12 h
3	3.0 times baseline OR increase in serum creatinine to ≥4.0 mg/dL (≥353.6 µmol/L) OR initiation of renal replacement therapy OR, in patients <18 years of age, decrease in eGFR to <35 mL/min per 1.73 m^2	<0.3 mL/kg per h for ≥24 h OR Anuria for ≥12 h

Abbreviation: eGFR, estimated glomerular filtration rate.

Harrison's Principles of Internal Medicine, 22nd Edition > Acute Kidney Injury
Dan Longo, Anthony Fauci, Dennis Kasper, Stephen Hauser, J. Larry Jameson, Joseph Loscalzo, Steven Holland, Carol Langford

Don’t need to memorize, it’s just here so you can understand that Cr level and urine output is not only used for diagnosis, but also for staging the severity of AKI.

Diagnostic Evaluation of AKI

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It's important to exclude prerenal and postrenal AKI because they're often quickly reversible.

Prerenal

- Signs of volume depletion (orthostatic hypotension, poor skin turgor) or heart failure (decreased CO, peripheral edema)
- BUN:Cr ratio >20 due to increased tubular reabsorption of urea
- Typically normal urinalysis
- Low urine sodium (<20 mEq/L) and FENa <1%

Postrenal

- Renal ultrasound to look for hydronephrosis
- Urinalysis may be normal or reflect underlying cause
- Urinary catheterization for suspected obstruction below the bladder
- Abdominal CT for nephrolithiasis
- FENa and urine sodium are not reliable

Prerenal AKI can have FENa <1% vs. >2% in ATN because there's a problem in the kidney itself, which impairs reabsorption.

In suspected postrenal AKI, look for signs of underlying cause in urinalysis → RBC casts in glomerulonephritis, WBCs in AIN and interstitial nephritis, crystalluria in nephrolithiasis, etc.

Diagnostic Evaluation of AKI

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Diagnosing Intrarenal AKI:

- Urinalysis results depending on disease (e.g., granular casts in ATN, WBCs in AIN, proteinuria and RBC casts in RPGN)
- Blood tests helpful for the diagnosis of glomerulonephritis and vasculitis: complement levels, antinuclear antibodies (ANAs), antineutrophil cytoplasmic antibodies (ANCAs), antiglomerular basement membrane (anti-GBM) antibodies, and cryoglobulins.

If the cause of AKI is not apparent based on the clinical context, physical exam, lab studies, and radiologic evaluation, kidney biopsy should be considered (provides definitive diagnosis).

Complications of AKI

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- Electrolyte imbalances:
  - Hyperkalemia → may be life-threatening due to risk of arrhythmia!
  - Hyperphosphatemia
  - Hypocalcemia
- Metabolic acidosis
- Uremia → symptoms may include nausea, asterixis (flapping tremor), platelet dysfunction, pericarditis, encephalopathy (seizures)
- Na+ and water retention → severe fluid retention may lead to edema and HTN, sometimes to pulmonary edema and hypoxemia. Loop diuretics like furosemide can be used in an initial trial. If patients fail to respond, dialysis is required for fluid removal

Hyperkalemia (serum K+ >5.5 mEq/L)

The accumulation of unfiltered toxins and loss of renal function may lead to uremia. Uremia literally means “urine in the blood.” The buildup of toxins can lead to symptoms like…

Management of AKI

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General considerations:

1. Identify and correct the underlying cause
2. Optimize hemodynamics through volume resuscitation and judicious use of vasopressors (diuretics should generally be reserved for patients with severe fluid retention, such as those with CHF or pulmonary edema)
3. Maintain mean arterial pressure >65 mmHg
4. Eliminate nephrotoxic agents if possible
5. Sufficient protein and calorie intake (20–30 kcal/kg per day) to avoid negative nitrogen balance
6. Initiation of dialysis when indicated

Identify and correct the underlying cause:

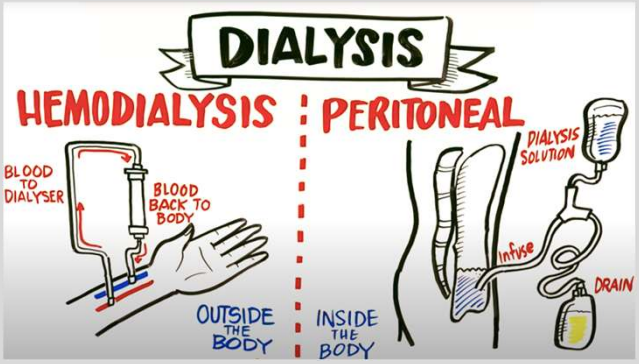
a. Prerenal → volume control. If volume depleted, prescribe isotonic saline. If volume overloaded (CHF), prescribe diuretics, inotropic agents, vasodilators (not including ACEIs)
b. Intrarenal → discontinue toxic medications, correct hypotension or infection, stabilize underlying disease (e.g., immunosuppressive therapy in RPGN, anticoagulation in acute renal embolus)
c. Postrenal → urgent relief of obstruction

nephrotoxic agents (e.g., ACE inhibitors, ARBs, NSAIDs, aminoglycosides, chemotherapeutic agents, checkpoint inhibitors, antibiotics)

Indications for Dialysis

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- Metabolic acidosis (pH<7.1)
- Severe or refractory hyperkalemia (>6.5 mEq/L)
- Severe azotemia (BUN>80-100)
- Fluid overload refractory to diuresis
- Uremic signs of pericarditis or encephalopathy
- Some alcohol or drug intoxications



Renal replacement therapy can be accomplished with peritoneal dialysis with a temporary intraperitoneal catheter, or hemodialysis using vascular access through the femoral, internal jugular, or subclavian veins.

Practice Question

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A 73-year-old man comes to the office due to 3 months of progressive urinary urgency, hesitancy, nocturia, and weak urinary stream. He has no fever, abdominal pain, hematuria, malaise, or weight loss. His only medication is lisinopril for hypertension. The patient has no history of diabetes or ischemic heart disease. He does not use tobacco, alcohol, or illicit drugs. Digital rectal exam reveals a smooth, firm, and enlarged prostate without induration or asymmetry. Neurological exam is normal. Urinalysis shows no proteinuria or hematuria. The patient's serum creatinine is 2.1 mg/dL, which is higher than his baseline creatinine of 1.2 mg/dL 4 months ago. Prostate-specific antigen (PSA) is normal. Which of the following is the most appropriate next step in evaluation of this patient's acute kidney injury?

A. Cystoscopy
B. Kidney biopsy
C. Prostate biopsy
D. Renal ultrasound
E. Urine cytology

Answer = D. Renal ultrasound

Based on symptoms and DRE, this patient has BPH with evidence of postrenal acute kidney injury. Cr is generally not elevated in unilateral urinary obstruction (eg, ureteral stone) but can be elevated in bilateral obstruction, such as in patients with sever bladder outlet obstruction due to BPH. Patients with acutely elevated Cr require imaging (preferably a renal US) to assess for hydronephrosis and exclude other causes of obstruction. Placement of a urinary catheter in patients with hydronephrosis can provide quick relief of the obstruction.

Summary

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Definition of AKI: Abrupt decline in GFR → ↑ serum creatinine or ↓ urine output

Etiologies:

Prerenal AKI is due to decreased perfusion of the kidney (ie, decreased renal blood flow or decreased glomerular capillary hydrostatic pressure, both of which are important in maintaining the GFR). There is no disease within the kidney itself so if renal perfusion improves early on, prerenal AKI can be reversible. Prerenal azotemia is the most common form of acute renal failure in the general population.

Intrarenal AKI occurs from structural derangement of the kidneys. The specific cause is determined by the *site of injury*: vasculature, glomeruli, tubules, interstitium.

In postrenal AKI, kidney function is impaired because the exit for urine is obstructed somewhere in lower urinary tract. This leads to high pressure that is transmitted into the tubules, leading to a decrease in the filtration pressure differential across the glomerulus. This will lower GFR. Common causes of post-renal AKI include BPH (most common), nephrolithiasis, neoplasm, ureteral clot/obstruction, trauma, vesicoureteral reflux in children.

Summary

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Diagnosis:

- KDIGO criteria (↑Cr ≥0.3 mg/dL in 48h or ≥1.5x baseline in 7d, or <0.5 mL/kg/h urine for 6h)
- Labs, urinalysis, imaging (e.g., renal US for obstruction)

Key Findings:

- Oliguria/anuria, volume overload, uremic symptoms
- Urine microscopy: muddy brown casts (ATN), WBCs (AIN), RBC casts (GN)

Complications: Hyperkalemia, acidosis, uremia, volume overload

Management:

- Correct underlying cause
- Avoid nephrotoxins
- Dialysis for severe/refractory complications

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